

RISK MANAGEMENT

Project # 3

Scenario

GWP3 is a continuation of the work you submitted for GWP1 and GWP2. That is, you are going to work towards the last two sections of your mini-capstone. Recall that each student individually reads the paper:

Alvi, Danish A. *Application of Probabilistic Graphical Models in Forecasting Crude Oil Price*. 2018. University College London, Dissertation.

<https://arxiv.org/abs/1804.10869>

By the end of GWP3, you will have a results and interpretation section. The results emphasize the findings of applying the methodology to the data, and assessing if the objective was fulfilled. The interpretation analyzes and evaluates the results and methodology.

When all GWPs are completed, the combined 3 projects look like a mini-capstone.

Overview of GWPs

GWP 1: Problem Formulation. Data Collection.

GWP 2: Methodology Description. Model Development.

GWP 3: Interpretation of Results. Improving the Model.

Tasks

Step 1

Defining the datasets.

- a. Student A writes 1–2 pages defining the purpose of the training set.
 - b. Student B writes 1–2 pages defining the purpose of the validation set.
 - c. Student C writes 1–2 pages defining the purpose of the testing set.
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Step 2

As a group, the team writes up to 6 pages comparing the validation and testing sets and shows the allocation of the data to training, validation, and testing.

Step 3

Validating the Model. Each person re-reads Section 4.3 of the dissertation.

Step 4

As a group, the team:

- a. runs “Re-running the Bayesian network using hill climbing”, and
- b. confirms whether they were able to replicate the results of the paper.

Step 5

Interpreting the results. As a group, the team:

- a. reports the accuracy of forecasting the price of crude oil, and
- b. develops a graphical way to display the results.

Step 6

Assessing the contributions. As a group, the team:

- a. assesses the 8 proposed results (from Danish paper). For example, “First, it contributes to the original research of replacing EGARCH-M derived views with Bayesian Model derived views for the Black-Litterman model...”;
- b. cites the specific pages, graphs, and results from the paper for each proposed result;
- c. reflects critically on whether the author’s work accomplished what was proposed;
- d. if the group feels the author achieved this contribution, the group then indicates if/why they think this is important.

Step 7

Application. As a group, the team writes a non-technical version of the results as the “Discussion” section. In particular, the section should focus on what this study does better, more quickly, or more cost-effectively than existing prediction models.
